Unit: Unit 6: Systems of Linear Equations and Inequalities | **Name:** ____________
Topic: Systems of Linear Inequalities by Graphing | **Date:** ____________

Homework (Habanero)

Q1. What is the region of the x-y plane that satisfies the inequality $y > 2x - 3$?
(A) $y \geq 2x - 3$
(B) $y < 2x - 3$
(C) $y > 2x + 3$
(D) $y < 2x + 3$
(E) $y = 2x - 3$

Q2. A doctor orders a dosage of x milligrams of a medication, and the patient's reaction is modeled by $y = 0.5x - 1$. The patient's reaction is within a safe range when $yleq2$. What is the maximum dosage?
(A) 6 milligrams
(B) 4 milligrams
(C) 8 milligrams
(D) 10 milligrams
(E) 12 milligrams

Q3. Which of the following inequalities represents the region below the line $y = x + 2$?
(A) $ygeqx + 2$
(B) $yleqx - 2$
(C) $y < x + 2$
(D) $y > x - 2$
(E) $y = x + 2$

Q4. What is the intersection of the regions defined by $xgeq0$ and $yleq3$?
(A) $x = 0$, $y = 3$
(B) $xgeq0$, $yleq3$
(C) $x < 0$, $y > 3$
(D) $x = 3$, $y = 0$
(E) $xleq3$, $ygeq0$

Q5. The growth rate of bacteria is modeled by the inequality $y > 2x + 1$, where y is the number of bacteria and x is the time in hours. What is the minimum time it takes for the bacteria to reach 10?
(A) 2 hours
(B) 3 hours
(C) 4 hours
(D) 5 hours
(E) 6 hours

Q6. A medical researcher is studying the effect of two medications on patients. The first medication has an effect modeled by $y = 0.2x + 1$, and the second medication has an effect modeled by $y = 0.3x - 2$. What is the point of intersection?
(A) (5, 2)
(B) (10, 3)
(C) (15, 4)
(D) (20, 5)
(E) (25, 6)

Q7. The inequality $y < 0.5x - 2$ represents which region of the x-y plane?
(A) Above the line
(B) Below the line
(C) On the line
(D) To the left of the line
(E) To the right of the line

Q8. A hospital is planning to build a new wing with a rectangular shape, where the length l and width w satisfy the inequality $l + wleq100$. What is the maximum area of the wing?
(A) 100 square meters
(B) 200 square meters
(C) 250 square meters
(D) 500 square meters
(E) 1000 square meters

Open Ended Questions (FRQ)

Q9. A pharmaceutical company is testing a new medication that has two active ingredients. The first ingredient has a concentration x and the second ingredient has a concentration y. The medication is effective when $x + ygeq5$ and $x - yleq2$. Graph the region of the x-y plane where the medication is effective.

Q10. A patient's body mass index (BMI) is calculated using the formula $BMI = \frac{weight}{height^2}$. The patient's BMI should be between 18.5 and 24.9. If the patient's height is 1.7 meters, what is the range of weights that satisfy the BMI requirements?

Chef's Tips

- Use graph paper to visualize the regions.
- Check for overlapping solution regions.
- Consider the context of the problem when interpreting the results.